

A small improvement of the seven-eighths zero-free boundary from a compensated probe

RH–Weil research project

October 7, 2026

A proof relative to explicitly cited arithmetic and analytic inputs

Abstract

We give a parameter extension of the compensated-probe argument in the September 30, 2026 version of *The Quasi-Riemann Hypothesis*. Relative to its specified generic arithmetic and analytic results and its finite-order Hecke seven-eighths theorem, we obtain the strict zero-free half-plane $\operatorname{Re} s > 69999/80000$ for finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$, and hence for all Dirichlet L -functions. The total prime-slot length is increased from $1/6$ to $1/6 + 1/20000$. A retained positive loss in the reflected row energy is paid by the rescaling coefficients. We check the enlarged Euler domain, the actual detector capacities, and the joint conductor cost of all ramified errors. An exact continuous inequality gives a uniform central saving $307/11016000$. We then specify the order of all choices, close the height truncation, and use Mellin continuation for the full character family. The assertion permits the principal pole at $s = 1$ and does not include the boundary line. This paper is a cited-input derivation; it is not an independent certification of the source or a Lean formalization of the improvement.

Contents

1	Statement, provenance, and imported results	1
2	The same compensated physical expression	3
3	The low bound at variable slot length	4
4	Euler corrections, contours, and the principal signal	5
4.1	Principal normalization	7
4.2	Whole buffered bins and the outer rows	7
5	Actual detector capacities and whole-slot counts	8
6	Joint errors and a continuous high-side saving	9
7	Parameter order, height closure, and continuation	11
A	Verification and reproducibility	13

1 Statement, provenance, and imported results

Put $F = \mathbb{Q}(\sqrt{-3})$ and $q_a = N(a)$, with the source convention for ideals, elements, ray representatives, and units. Characters are extended by zero on nonunits. The superscript S deletes the

finite Euler factors at the fixed set S . All assertions about a principal L -function allow its pole at 1.

Our source is the September 30 preprint [1], at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its canonical UTF-8 SHA-256, after replacing CRLF and lone CR by LF, is recorded in Appendix A. The line references below refer to that exact source, rather than to a moving branch.

Definition 1.1 (Imported input package $\mathcal{R}$). The package $\mathcal{R}$ consists of the following results with their stated uniformities and hypotheses in [1]: the coefficientwise probe and Poisson identities and local arithmetic tables; smooth separation, Gaussian annuli, reflection sectors and generic reflected energy; the generic additive Gram bound; the global and buffered reciprocal estimates, Hecke growth and functional equation; the detector witnesses, marked and plain moments, sixth-power inverse amplification, and uniform slot mesh; the fixed-ray prime asymptotic; and the source theorem for all primitive finite-order Hecke characters with strict boundary $7/8$. The locations and the uses of these inputs are given in Table 1. Their correctness is assumed here.

Theorem 1.2 (Boundary improvement relative to $\mathcal{R}$). *Assume $\mathcal{R}$. For every finite-order Hecke character η of F , including imprimitive characters,*

$$L_F(s, \eta) \neq 0 \quad \left(\operatorname{Re} s > \frac{69999}{80000} \right).$$

For every Dirichlet character χ the same assertion holds for $L(s, \chi)$, and in particular for $\zeta(s)$. The principal pole at $s = 1$ is allowed. No assertion is made about zeros on the line $\operatorname{Re} s = 69999/80000$.

This is a change of $1/80000$ in the boundary of the cited theorem. The proof does not use a proportion of zeros on the critical line. It makes no claim about the Riemann hypothesis or the novelty of the generic inputs.

Table 1: Exact imported inputs and the new checks performed here.
Source labels and line intervals refer to [1].

Input	Source location	Use and retained hypotheses
Smooth calculus and Gaussian annuli	<code>lem:smooth-calculus</code> , <code>lem:gaussian-annular</code> ; 1123–1209, 1288–1315	Fixed finite-dimensional common Fourier profile, weighted seminorms, and whole annular tails; no restriction to slot length $1/6$.
Reflection and energy	<code>lem:reflection-row-sectors</code> , <code>lem:reflected-energy</code> ; 2470–2506, 7663–7804	Actual row masks, powerful fixed parts, retained dual support and bounded logarithmic lengths. Section 3 pays the changed loss.
Additive Gram	<code>prop:probe-gram</code> ; 8340–8364	Actual ray/Gauss coefficients, fixed polynomial ranges $Q, Y' \geq 1$, and $P_a = Y'^2/Q \geq 1$.
Physical high identity	<code>def:stage-high-data</code> ; 5582–5649; local compensation 8707–9049	An independently defined completed probe with finite compensation. Section 4 proves its new-domain correction without dividing by selected factors.
Reciprocal and contours	400–502, 5672–5728, 5808–5896, 6021–6155	Whole buffered bins, all-height outer tails, and fixed real boxes. No contour crosses a target zero.
Principal and outer rows	6174–6301, 6342–6463, 9101–9195	Same finite S , same principal normalizer; independent absolute bounds for small and large rows.

Input	Source location	Use and retained hypotheses
Detectors and moments	lem:detector-dyads ; 4385–4547; marked 9220–9269; inverse amplification 12362–12389; plain 12492–12578	Actual witnesses, both common signs, allowed heights and capacities, and $\kappa \geq 3/4$. Section 5 checks each restriction.
Actual slot counts	15015–15446	Whole slots, natural coefficients, inducing-character exclusion, fixed finite presentation set, rounding, and common mesh.
Bootstrap and transfer	16455–16463; 6705–6780	Whole Hecke 7/8 theorem used only to bound β_* from above; quadratic transfer applied after the improved Hecke conclusion.
Height and order	6520–6580, 16191–16453	Internal derivative orders fixed before the external tail order; target-dependent height exponent chosen last.

2 The same compensated physical expression

The arithmetic data S, T, b_*, ξ , completed-index convention $A = cn^3$, Gaussian test, ray calibration, and all zero-on-nonunit masks are those of the source. Write $I_\eta(X, Y, Z)$ for its completed probe (**eq:general-probe**), and $I_\eta^{[p|A]}$ for the same expression restricted to $p \mid A$. The completed sums are independently defined and absolutely convergent on the initial Mellin line; their outer annular sums and the compensation over prime tuples and subsets are finite. They are not defined by a high integral.

For a fixed prime tuple the two operations in **eq:compensated-probe-definition** are the marked operation $\overline{\eta(p)} I_\eta^{[p|A]}(X, Y, Z q_p)$ and the rescaled operation $q_p^{-3/2} I_\eta(X/q_p, Y/q_p, Z)$, with a minus sign between them. Apply these operations in each of the disjoint slots, multiply by $W_i(q_p/P_i)$, and sum over the tuples. Denote the resulting finite combination by $I_{\eta, \text{mod}}(Z)$. The inherited restrictions and calibration remain part of this definition. More explicitly, if $p_J = \prod_{i \in J} p_i$, $p_\emptyset = 1$, and $I_{\eta; D}$ inserts $1_{D|cn^3}$ in the completed row, this is

$$I_{\eta, \text{mod}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_J^c)} \cdot I_{\eta; p_J^c}(X/q_{p_J}, Y/q_{p_J}, Z q_{p_J^c}). \quad (1)$$

Every window stays at its original scale P_i , including when another slot changes the completed scale.

We use the following fixed geometry:

$$\begin{aligned} b &= \frac{1}{8}, \quad \ell = \frac{10003}{60000} = \frac{1}{6} + \frac{1}{20000}, \quad l_x = \frac{1 - \ell - b}{2} = \frac{42497}{120000}, \\ l_y &= \frac{1 - \ell + b}{2} = \frac{57497}{120000}, \quad M = 1 - \ell, \quad h = 1 - l_x + \ell = \frac{32503}{40000}, \\ X &= Z^{l_x}, \quad Y = Z^{l_y}, \quad \sigma_1 = \frac{69999}{80000}, \quad C(s) = s + \frac{l_x}{2} - 1 + \frac{h}{6} = s - \frac{11}{16}. \end{aligned} \quad (2)$$

In particular $C(\sigma_1) = 14999/80000$. Take a fixed even integer K , positive equal lengths $\ell_i = \ell/K$, and $P_i = Z^{\ell_i}$. Choose nonnegative nonzero smooth W_i with compact support in the pairwise disjoint intervals

$$\left(1 + \frac{2i-1}{2K+1}, 1 + \frac{2i}{2K+1}\right) \subset (1, 2).$$

The slots are the allowed source prime representatives in T with q_p/P_i in these supports. Their disjointness makes every selected prime occur once.

3 The low bound at variable slot length

Proposition 3.1. *Under the generic low inputs in $\mathcal{R}$, for every fixed $1/6 \leq \ell \leq 1/5$, $b = 1/8$, and the corresponding geometry above,*

$$|I_{\eta, \text{mod}}(Z)| \ll_{\eta, \text{fixed}, \varepsilon} Z^{(1-\ell)/4-b/6+\varepsilon}.$$

The exponent loss $\varepsilon > 0$ can be prescribed before the target; constants and thresholds may depend on its fixed arithmetic data.

Proof. Fix a rescaled subset J and its prime tuple, and put $d_J = \sum_{i \in J} \ell_i$, $\ell' = \ell - d_J$, $M' = M - 2d_J$. The norm ratio $r_J = q_{p_J}/Z^{d_J}$ lies in a fixed compact subinterval of $(0, \infty)$. The exact low separation has a common row ball

$$X' = XZ^{-d_J}/r_J, \quad Y' = YZ^{-d_J}/r_J, \quad Q = q_{b_*}X'Y' \asymp Z^{M'}.$$

All surviving marked slots are summed inside B_m^J before the row norm is taken. The completed index remains cn^3 , with weight $q_c^{-1/2}q_n^{-1}$, including permitted shared primes.

Partition the Gaussian in $q_{cn^3}/Z^{1+\ell'}$ into whole annuli. On $\log(q_{cn^3}/Z^{1+\ell'}) = k + O(1)$ the common profile is

$$\chi(\log x) W_G \left(e^k x / \prod_i \varrho_i \right) \prod_i W_i(\varrho_i).$$

Its weighted Fourier seminorms are summable over k . First use the absolute completed-sum bound and Gaussian decay to remove $|k| > \kappa_G \log Z + O(1)$ with arbitrary power saving. The retained norm exponent is $N_* = 1 + \ell' + \theta_N$, $|\theta_N| \leq \kappa_G + O(1/\log Z)$. Fix this common Fourier density before separating rows or tuples. The separated slot coefficients retain their arithmetic phases, annular cutoffs and unit-modulus norm powers, so the generic reflected-energy lemma applies.

Freeze a powerful row part of logarithmic length O and an actual residual squarefree dyad of length H . The source sectors give

$$\Delta_H = M' - O - H \geq -\varepsilon_Z, \quad 2A_0 \leq O + \varepsilon_Z, \quad N_0 \leq A_0, \quad z_a \leq \ell', \quad \varepsilon_Z = O_{\text{fixed}}(1/\log Z).$$

The dual length is $T_d = 2H + 2A_0 + 2z_a - 1 - \ell' - \theta_N - N_0 - 3B_0$. Since $M' + \ell' - 1 = -3d_J$,

$$T_d - (H - 3d_J) = H - M' + 2A_0 + 2(z_a - \ell') - N_0 - 3B_0 - \theta_N \leq 2\varepsilon_Z + |\theta_N|. \quad (3)$$

This calculation retains the original support and shared S -primes. For completeness the imported energy expression, after counting the frozen parts once, is

$$s_{\text{hyb}} = \min\{v, z_a, (v + z_a)/3\}, \quad y = v + 3\ell_b + e_\lambda \leq T_d + \tau_{\text{ref}}, \\ E_{\text{ref}} = O/2 + \max(H, v + \ell_b) - S_0 - B_0 + z_a - s_{\text{hyb}} - \ell_b - 2e_\lambda/3 - (T_d - y)_+/2.$$

Here v, ℓ_b are the nonnegative dual ideal lengths, S_0, B_0, N_0, A_0 are the frozen-part lengths of the source sectors, and a negative e_λ is only $O(1/\log Z)$. The retained support and (3) imply $\max(H, v + \ell_b) = H + O(\tau_{\text{ref}} + \kappa_G + \varepsilon_Z)$. When $s_{\text{hyb}} = z_a$, cancel $z_a - s_{\text{hyb}}$ and discard the other nonpositive terms: this gives at most $M' - \Delta_H$ plus small losses. Otherwise $s_{\text{hyb}} \geq v/2$, and

$$s_{\text{hyb}} + \ell_b + 2e_\lambda/3 + (T_d - y)_+/2 \geq y/4 + (T_d - y)_+/2 - O(\varepsilon_Z) \geq T_d/4 - O(\varepsilon_Z).$$

The exact saving identity is

$$O/2 + \Delta_H + S_0 + B_0 + T_d/4 = \frac{2M' + 2z_a - 1 - \ell' + 2\Delta_H - \theta_N}{4} \\ + \frac{2A_0 - N_0 + B_0 + 4S_0}{4}.$$

The second line is nonnegative. Using $z_a \leq \ell'$ gives the second branch bound

$$M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H + \theta_N}{4} + O(\varepsilon_{\text{ref}} + \kappa_G + \varepsilon_Z).$$

Here $1 + 3\ell' - 2M' = 5\ell - 1 + d_J$. Both branches decrease with Δ_H . Extract the kernel saving from the common profile before squaring or positive enlargement. Fourier Minkowski, the annular sum, and the finite sector/dyadic sums therefore give

$$\sum_{0 < q_m \ll Q} |B_m^J|^2 \ll Z^{M' + ((5\ell - 1 + d_J)/4)_+ + \varepsilon}. \quad (4)$$

The powerful-part count is already included here. It must not be multiplied in a second time. Empty surviving slots lie in the first branch, apart from vanishing bounded logarithmic offsets.

Throughout $1/6 \leq \ell \leq 1/5$ we have

$$M' \geq 2/5, \quad l_x - d_J \geq 11/80, \quad l_y - d_J - 11b/6 \geq 1/30.$$

Thus $Q, Y' \geq 1$ eventually and are in fixed polynomial ranges. Also $P_a = Y'^2/Q = q_{b_*}^{-1} Z^b \geq 1$ and $P_a^2/Y' \ll P_a^{1/6}$. The generic additive Gram result yields

$$\sum_{q_m \ll Q} |A_m(Y')|^2 \ll (1 + |\nu|)^{J_{\text{Gr}}}(Q/Y') P_a^{1/6} Z^\varepsilon.$$

The original Mellin test absorbs this height power. Cauchy–Schwarz on the same ball, including its $Q^{-1/2}$ normalization, bounds a fixed rescaled tuple by

$$(X')^{1/2} P_a^{1/12} Z^{(5\ell - 1 + d_J)_+ / 8 + \varepsilon}.$$

There are at most $Z^{d_J + \varepsilon}$ such tuples; their outer coefficients are $O(Z^{-3d_J/2})$, and shortening X' contributes $Z^{-d_J/2}$. The total extra exponent is

$$-d_J + \frac{(5\ell - 1 + d_J)_+}{8} \leq -\frac{7d_J}{8} \leq 0,$$

because $5\ell - 1 \leq 0$. This pays, rather than deletes, the positive loss in (4). Sum the finitely many subsets to obtain $X^{1/2} Z^{b/12 + \varepsilon} = Z^{(1-\ell)/4 - b/6 + \varepsilon}$. Choose the reflection, separation and annular losses within the prescribed ε , then fix seminorm orders, and finally raise the threshold. All logarithmic multiplicities are absorbed at this last step. $\square$

For (2) the low exponent equals $C(\sigma_1)$ exactly.

4 Euler corrections, contours, and the principal signal

Let V, W, D, P_p, P_p^*, H_p be the source local factors; W here is a local Euler factor, distinct from the smooth slot windows. For a sixth-power-free physical row u , write $\chi_\bullet(u)$ for its source character presentation. More explicitly, with $Q = q_p$ and $j = v_p(u) \in \{0, \dots, 5\}$, put

$$\begin{aligned} \rho &= \chi_p(u/p^j), \quad a_p = \overline{\alpha(p)}^3 \eta(p)^3, \quad b_p = \overline{\alpha(p)}^2 \eta(p)^2 \overline{\chi_p(4)}, \\ V &= Q^{-6z}, \quad R = a_p^2 Q^{4-6s-6z}, \quad W_{\text{loc}} = \rho Q^{-w}, \quad W = \chi_p(u) Q^{-w}, \quad D = \eta(p) \overline{\chi_p(u)} Q^{-s}. \end{aligned}$$

The source phases have modulus one and $b_p^3 = a_p^2$. Its coefficientwise local table gives

$$\begin{aligned} P_p^* &= \frac{1}{1-R} \left\{ \frac{R(1-Q^{-1}) - \eta(p)(Q-1)Q^{-s-w}V^{1_{j \leq 1}}}{1-V} + J_j \right\}, \\ P_p &= (1-V)^{-1} + 1_{j=0} \frac{W_{\text{loc}}}{1-W_{\text{loc}}} + P_p^*, \quad H_p = P_p \frac{(1-V)(1-W)}{1-D}, \end{aligned}$$

where

j	J_j
0	$-\eta(p)\rho^{-1}Q^{-s} + W_{\text{loc}}R$
1	$\eta(p)Q^{-s-w}$
2	$a_p\rho^{-3}Q^{3/2-3s}$
3	$-\eta(p)b_p\rho^{-2}Q^{2-3s-w} + b_p^2\rho^{-4}Q^{2-4s}$
4	$-\eta(p)a_p\rho^{-3}Q^{5/2-4s-w}$
5	$-a_p^2Q^{3-6s}$.

At $p \mid u$, $D = W = 0$, whereas W_{loc} retains its phase. The exact local identity is

$$H_p - 1 = \frac{D(V + W - VW) - VW + (1 - V)(1 - W)E_p}{1 - D}, \quad E_p = P_p^* + D. \quad (5)$$

In particular this expression has no $1 - W$ denominator. Use the domains

$$\begin{aligned} \mathcal{D}_1(\varepsilon_0) &= \{s_r \geq 51/100, z_r \geq 17/50, w_r \geq -1/100, s_r + w_r \geq 1 + \varepsilon_0\}, \\ \mathcal{D}_2^* &= \{s_r \geq \sigma_1, z_r \geq 33/200, w_r \geq 19/20\}, \end{aligned}$$

where subscripts r denote real parts. The source local table gives, at good primes, $E_p \ll Q^{4-6s_r-6z_r+\theta} + Q^{1-s_r-w_r-6z_r}$, where $\theta = (-w_r)_+$ and $Q = q_p$. On $\mathcal{D}_2^*$, $\theta = 0$. Substitution in (5) gives on $\mathcal{D}_2^*$

$$H_p - 1 \ll Q^{-1-c_b}, \quad c_b = \min(6\sigma_1 - 401/100, \sigma_1 - 3/50, 47/50) = 65199/80000.$$

At ramified primes it gives $H_p - 1 \ll Q^{-c_r}$, where $c_r = \min(\sigma_1 - 1/20, 3\sigma_1 - 3/2) = 65999/80000$. On $\mathcal{D}_1(\varepsilon_0)$ the corresponding exponents are $-1 - \varepsilon_H$ and $-\varepsilon_H$, where $\varepsilon_H = \min(\varepsilon_0, 1/50) > 0$. Strict positive margins yield normal convergence on neighborhoods, uniform in all imaginary parts, with ramified finite product q_u^ε .

Define the selected local replacement by

$$G_p = \frac{(\overline{\eta(p)}Q^s - Q^{-w})(1 - V)(1 - W)P_p^* - Q^{-w}(1 - VW)}{1 - D}.$$

Define the full correction by the finite sum

$$\mathfrak{H}_{\eta,u,Z} = \sum_{(p_i)} \prod_i \{W_i(q_{p_i}/P_i)q_{p_i}^{z-1}G_{p_i}\} \prod_{\substack{p \notin S \\ p \notin \{p_1, \dots, p_K\}}} H_p. \quad (6)$$

This definition does not divide by P_p or H_p . The local table bounds $G_p \ll Q^B$ on every fixed real box, uniformly in heights. Counting the tuples gives $\mathfrak{H}_{\eta,u,Z} \ll q_u^\varepsilon Z^{B'}$, with B' fixed before any external integration-by-parts order.

On the absolute starting lines the exact identity

$$\overline{\eta(p)}Q^{s+z-1}P_p^* - Q^{z-w-1}P_p = \frac{1 - D}{(1 - V)(1 - W)}Q^{z-1}G_p$$

shows that the finite operation of Section 2 extracts the original scalar quotient, once per selected prime. Consequently it proves, for that independently defined physical sum,

$$\begin{aligned} I_{\eta,\text{mod}}(Z) &= \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} \mathcal{W} \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \\ &\quad \frac{\zeta_F^S(6z)L^S(w, \chi_\bullet(u))}{L^S(s, \eta\chi_\bullet(u))} \mathfrak{H}_{\eta,u,Z}(s, w, z) dz dw ds, \end{aligned} \quad (7)$$

where all physical row restrictions are retained, and

$$\mathcal{W} = X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} M(z) \widehat{W}_1(w).$$

The source coefficientwise identity supplies absolute convergence on these starting lines. The new local estimates supply its required analytic domain and all-height majorant.

4.1 Principal normalization

Choose P_0 , before choosing a target, so that $\exp(C_0 \sum_{Q > P_0} Q^{-1-c_b}) - 1 \leq 1/2$. Include these small primes in S . Enlarging S later only shrinks this positive majorant. For $u = 1$ the unselected product $H_\eta(s) = \prod_{p \notin S} H_p(s, 1, 1/6)$ is holomorphic on $\text{Re } s > \sigma_1$ and $|H_\eta - 1| \leq 1/2$ there. Only for this principal row may we now use G_p/H_p throughout $\mathcal{D}_2^*$. The four local error powers are

$$-s_r, \quad -6z_r, \quad 4 - 5s_r - 6z_r, \quad 1 - w_r - 6z_r.$$

Their maxima on this domain are $-\sigma_1, -99/100, -21839/16000, -94/100$, respectively. Thus $G_p/H_p = -1 + O(Q^{-\sigma_1})$.

Let $S_i(Z) = \sum_p W_i(q_p/P_i) q_p^{-5/6}$ in the actual ray. The imported fixed-ray prime asymptotic gives $S_i \sim c_i P_i^{1/6} / \log P_i$, $c_i > 0$. For our even K set

$$A_T(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z).$$

It is eventually nonzero, $A_T \asymp (\log Z)^{-K}$, and $A_T^{-1} \ll_\varepsilon Z^\varepsilon$. For every fixed $0 < \mu < \sigma_1 \min_i \ell_i$,

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_T(Z) (1 + R_{\eta,Z}(s)), \quad R_{\eta,Z}(s) \ll Z^{-\mu}. \quad (8)$$

The bounds are uniform in heights on the global principal line. Put

$$c_S = \widehat{W}_1(1) M(1/6) \frac{(\text{Res}_{s=1} \zeta_F^S(s))^2}{6} > 0.$$

Moving the principal w, z lines to $1 + e, 1/6 + e$, then $w = 19/20$ across 1 and $z = 33/200$ across $1/6$, stays in $\mathcal{D}_2^*$. Keep $s = \beta_* + e$ throughout these moves. The reciprocal is globally controlled there; it crosses no zero. Only the main double residue is moved right to $s = 2$. Define

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds. \quad (9)$$

If P_η denotes the full principal row, the residual integrals, including full two- and three-dimensional tails, satisfy

$$|P_\eta / (c_S A_T) - f_\eta| \ll Z^{C(\beta_*) + (1+h)e - m_w + \varepsilon} + Z^{C(\beta_*) + e - m_z + \varepsilon} + Z^{C(\beta_*) + e - \mu + \varepsilon}, \quad (10)$$

$$m_w = 57497/2400000, \quad m_z = 32503/24000000.$$

4.2 Whole buffered bins and the outer rows

Define

$$\beta_* = \sup(\{1/2\} \cup \{\text{Re } \rho : L_F(\rho, \eta) = 0, 1/2 \leq \text{Re } \rho \leq 1, \eta \text{ primitive of finite order}\}).$$

Poles are excluded from this set. Assume for contradiction $\sigma_1 < \beta_* \leq 7/8$, with the upper bound supplied by $\mathcal{R}$. Use the original buffered-bin partition before moving any row contour. Its floor is $a = 51/100$; elsewhere $a \leq \beta_*$. At $z_0 = 17/50$ move the whole bin first to $s = \beta_* + 20e$, $w = 1 - a - 6e$. Remove the outer height box on this global line, then move the retained bin to $s = a + 16e$. We have $s + w \geq 1 + 10e$ and $s > a + 6e$, so $\mathcal{D}_1(10e)$ and the buffered reciprocal bound apply. The floor need not have a witness. Any pointwise amplitude or error partition is made after this whole-bin move. Discarded pieces and joins are $O_{\eta,N}(Z^{B_\eta} T_1^{-N})$ with B_η fixed before N .

Here are independent controls for the outer rows, needed to cover every physical row. Put $d_{\min} = 1/100$. On the small-row lines $(s, w, z) = (\beta_* + e, 1/2, z_0)$ use $\mathcal{D}_1(1/3)$: $\sigma_1 + 1/2 > 1 + 1/3$. The old domain $\mathcal{D}_1(3/8)$ would be insufficient. The full local table gives $G_p = O(1)$ off the row and $G_p \ll Q^{1/2}$ on it. Thus each positive selected-slot sum is at most $q_u^\varepsilon P_i^{z_0}$, without dividing by H_p . Hecke growth contributes $U^{3/5}$, row count U , and $q_u^{-z_0}$ contributes U^{-z_0} , giving $U^{63/50}$. The exponent relative to $C(\beta_*)$ for $U \leq Z^{d_{\min}}$ is at most

$$h \frac{13}{75} - \frac{l_y}{2} + \frac{63}{50} d_{\min} = -\frac{172249}{2000000}. \quad (11)$$

For $U > Z^{h+\zeta}$ keep the full tuple and use absolute lines $(s, w, z) = (2, 2, z_\infty)$, $z_\infty > 2$. The bound is

$$Z^{B_0 + (h+\zeta)(1+\varepsilon) - \zeta z_\infty + \varepsilon}, \quad B_0 = \frac{132497}{80000}. \quad (12)$$

For any fixed $\zeta > 0$, choose z_∞ sufficiently large before the target. It gives any prescribed fixed saving. These estimates retain all bounded units and the full absolute row tails.

5 Actual detector capacities and whole-slot counts

The plain moment input requires $3/4 \leq \kappa \leq 1$ and, if $\kappa < 1$, $\beta_* \leq (1 + \kappa)/2$. Under the contradiction above we may fix $\kappa = 3/4$. In general the admitted value would be $\max(3/4, 2\beta_* - 1)$, with excess $\max(\beta_* - 7/8, 0)$. Here that excess is zero.

Write $\delta = 2a - 1$. For positive amplitude bins use the source bounds $|Q_i| \leq P_i^{g_i + \vartheta}$ and, only if $g_i > 0$, $|Q_i| \geq P_i^{g_i}$, with a prescribed small bin loss $\vartheta > 0$. Set $g_i = 0$ for error slots and for main slots with $|Q_i| \leq 1$; the latter need only an upper bound and request no positive slot. Absorb the preliminary bin losses separately, with $0 \leq g_i \leq \delta/2$, and let $q = \sum_i \ell_i g_i / \ell \in [0, \delta/2]$ be their length-weighted mean; put $x = q/\delta \in [0, 1/2]$. Thus a squared slot product has gain exponent $2q$. The inverse amplification is on the actual injective map $(u, a') \mapsto u(a')^6$, with conductor $\ll U$ and width one. It gives $e(r) = \max(1, (1 + 5r)/6)$, hence the long count

$$L(t) = 1 - \delta + (5/6 - \delta)(t - 1). \quad (13)$$

For an actual detector witness the imported dyadic construction gives

$$r + m \geq t - o(1), \quad r \leq t + o(1), \quad t - 1/2 - O(\varepsilon) \leq r, \quad 0 \leq m \leq 1/2 + O(\varepsilon),$$

with $|M_r|^2 \gg U^{\delta r - \varepsilon}$ and $|S_m|^2 \gg U^{\delta m - \varepsilon}$. The two occurrences of S_m in the plain argument give $|S_m|^4$. Its slot capacity is $(1 - 2m)/(6\kappa)$; inverse amplification has capacity $(1 - r)/2$.

In the inverse branch $r \geq r_*$, for a positive requested slot length $z = (1 - r)/2 - \nu_0$, decreased by a fixed $\nu_0 > 0$, the inverse capacity inequalities include

$$1 - r - 2z \geq 2\nu_0, \quad 3 - 2r - 8z \geq 8\nu_0 + 9/37 - O(\varepsilon).$$

In the plain branch $r \leq r_*$, use $z = (1 - 2m)/(6\kappa) - \nu_0$; its inequality is $1 - 2m - 6\kappa z \geq 6\kappa\nu_0$. The ideal short exponents at $\kappa = 3/4$ are

$$A_I(r) = 1 - \delta[x + (1 - x)r], \quad S_t(r) = 1 - \delta[4x/9 + (2 - 8x/9)(t - r)].$$

Set

$$D_x = 3 - 17x/9, \quad P_x = (2 - 8x/9)(1 - x), \quad J_x = (5/6 - \delta)D_x + \delta P_x. \quad (14)$$

Their crossing is $r_* = [(2 - 8x/9)t - 5x/9]/D_x$ and gives

$$R_{\text{short}} = 1 - \delta + \delta(P_x/D_x)(3/2 - t).$$

On $1 \leq t \leq 3/2$, $r_* \geq 23/37$, $1/3 \leq t - r_* \leq 1/2$, and $r_*(3/2) = 1$. The maximal inverse and plain capacities are respectively $7/37$ and $2/27$. Near saturated $r \geq 1$ use amplification; near $m = 1/2$ use the zero-slot plain moment. A capacity at most ν_0 costs at most $2\delta\nu_0$ or $6\delta\nu_0$, respectively, in the corresponding count.

The actual normalized prime factors are

$$Q_i(u; z) = P_i^{-1/2} \sum_{p \in \mathcal{P}_i(Z)} \overline{\chi_p(u)} W_i(q_p/P_i) (q_p/P_i)^{z-1}. \quad (15)$$

Primes dividing u have zero character value. The fixed presentation is $\psi = \overline{\nu\chi_\bullet(u)}$, with $\nu \in \Theta = \langle \eta, \widehat{T} \rangle$, a fixed finite group for the target. Its row-independent prime coefficients are $\overline{\nu(p)}1_T = |T|^{-1} \sum_{\theta \in \widehat{T}} (\bar{\nu}\theta)(p)$, a finite combination of members of Θ . In a plain moment, conjugate both witnesses and all selected factors; the coefficients become $\nu(p)1_T$. The supports and masks are unchanged.

The source inducing-character exclusion also remains valid: at a prime with sixth-free ramification label $j = 1, \dots, 5$, the local order $6/\gcd(6, j) > 1$ cannot be cancelled by an unramified fixed presentation. The remaining S -supported units form a finite set, eventually outside selected rows $U \geq Z^{1/2}$; outer estimates retain them.

For $U = Z^d$ the whole-slot lengths are $w_i = \ell_i/d$. Greedy selection of full positive slots loses at most one fraction, so its gain satisfies $|\prod Q_i|^2 \geq U^{2qz - \delta \max_i w_i}$. If the positive supply is insufficient, taking it all still gives gain at least qz . For the selected range $d \geq 1/2$, $w_i \leq 2\ell_i$. There is strict supply:

$$\frac{\ell}{h} - \frac{7}{37} = \frac{57659}{3607833} > 0, \quad 0 < \zeta < 5\ell - h = \frac{2521}{120000} \implies \frac{\ell}{h + \zeta} > \frac{1}{5} > \frac{7}{37}.$$

The latter gap is $2/185$. Choose the uniform plain-moment mesh before K , and then fix even K with $2\ell/K < \min(\text{mesh}, \text{rounding allowance}, 1/185)$. The actual w_i are smaller than this mesh, and every physical slot norm is at most $2U^{\text{mesh}}$. The plain moment's internal pool starts at $U^{\text{allowed width}/3}/2$. Choose mesh $< \text{allowed width}/6$, giving a strict exponent gap and eventual disjointness from that pool. Annular offsets are $O_K(1/\log U)$ and enter only after all strict margins have been fixed.

The admitted marked, inverse and plain moments now count the actual rows, with the original witnesses and whole selected slots:

$$\#\mathcal{B} \ll U^{\max(R_{\text{short}}, L(t)) + \varepsilon} (1 + T_1)^A.$$

Take

$$t = 1 + \frac{\delta P_x}{2J_x}, \quad R_* = 1 - \delta + \frac{(5/6 - \delta)\delta P_x}{2J_x}. \quad (16)$$

Here $35/54 \leq J_x \leq 5/2$, $D_x \geq 37/18$, $P_x \geq 7/9$, and $1 < t < 3/2$. This balances the two counts and proves $\#\mathcal{B} \ll U^{R_* + \varepsilon} (1 + T_1)^A$. The no-slot choice $t = 1$ instead gives exponent $1 - 2\delta/3$. The floor uses the trivial exponent 1, without an invented witness.

6 Joint errors and a continuous high-side saving

On the retained dynamic line $(s, w, z) = (a + 16e, 1 - a - 6e, z_0)$, the local product satisfies good/ramified bounds Q^{-1-10e} and Q^{-10e} . After fixing e , enlarge the target-independent small-prime cutoff. Only on this dynamic line may we factor by H_p and expand

$$\mathfrak{H} = \mathcal{H}_{\eta, u} \sum_I \prod_{i \in I} D_i \prod_{i \notin I} (-P_i^{z-1/2} Q_i),$$

where D_i denotes the local error sum. The global contour argument used the full expression (6). Off the row its four error exponents are $-a - 4e$, $-2.04 + 12e$, $1.96 - 5a - 68e$, $a - 2.04 + 12e$, all at most -0.51 for sufficiently small e . At a ramified prime the normalized table has, at $e = 0$, powers

$$a - 2, \quad 1/2 - 2a, \quad (-a, 1 - 3a), \quad 1/2 - 2a, \quad 2 - 5a,$$

with e -coefficients $6, -32, (-26, -48), -42, -80$. These are strictly less than $-1/2$, except for the strict $j \geq 2$ term whose bound is Q^{-w_r} .

All such strict terms must share one conductor budget. For the entire triangle-expanded set of distinct strict primes, the primitive conductor satisfies $q_{f_u} \ll_S q_u \prod_p Q_p^{-(j_p-1)}$. Apply the numerator functional equation once, with $A^* = 1/2 - w_r = a - 1/2 + 6e$. Its reflected primitive value is at $a + 6e$ and is covered by the buffered reciprocal control, including the conjugate presentation. Deleted factors cost $U^{6e+\varepsilon}$. Each strict prime then contributes at most

$$Q_p^{z_0 - w_r - (j_p-1)A^*} \leq Q_p^{z_0 - 1/2}, \quad j_p \geq 2.$$

Restoring its deleted strict factor costs 1. Consequently, for every error subset I , including the empty one,

$$\left| L^S(w, \chi_\bullet(u)) \prod_{i \in I} D_i \right| \ll U^{\delta/2 + O(e) + \varepsilon} (1 + T_1)^A \prod_{i \in I} P_i^{z_0 - 1/2 + O(e) + \varepsilon}.$$

Assign gain $g_i = 0$ to errors and $0 \leq g_i \leq \delta/2$ to main factors, and let $q = \sum_i \ell_i g_i / \ell$. The joint pointwise exponent is therefore $U^{\delta/2 + \text{loss}} Z^{\ell(z_0 - 1/2) + q\ell + \text{loss}}$. The count of Section 5 uses this same q over all slots. No second conductor saving is charged.

After the source contour accounting, the exponent relative to $C(\sigma)$, before small losses and height powers, is

$$E_\sigma(d) = K_\sigma + (1/2 + \ell)\delta + \ell q - h(1 - R) + (d - h)(R + \delta/2 - z_0), \quad K_\sigma = 2/3 + \ell + b/6 - \sigma. \quad (17)$$

Equivalently the unfactored accounting is $a - \sigma + h(z_0 - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - z_0)$. The contour's explicit additional loss is $(16 - 6l_y)e + (1 + d)\varepsilon_c + d\varepsilon_d + \varepsilon_p$, besides previously fixed count, normalization and height losses.

Lemma 6.1 (Continuous exact certificate). *At the geometry (2), with $R = R_*$ and $q = \delta x$, $0 \leq x \leq 1/2$, $1/50 \leq \delta \leq 3/4$,*

$$E_{\sigma_1}(h) \leq -\frac{307}{11016000} =: -m_{\text{ad}}.$$

Proof. First set $\ell_0 = 1/6$, $\sigma_0 = 7/8$, $h_0 = 13/16$, and denote the endpoint exponent by E_0 . Put $y = 1/2 - x$, $v = 51 + 41y$, and use J_x of (14). Direct expansion gives the polynomial identity

$$\begin{aligned} 10368vJ_x(-E_0) &= (3 + 5y)\{(4v\delta - 79)^2 + 49\} \\ &\quad + 4y\{4v\delta[(1 + 3y)(15 + 32y)\delta + 9 - 13y] + 265 + 3485y\}. \end{aligned}$$

Every term is nonnegative for $0 \leq y \leq 1/2$ and $0 \leq \delta \leq 3/4$, since $9 - 13y \geq 5/2$. Also $v \leq 17(3 + 5y)$ and $J_x \leq 5/2$, hence $-E_0 \geq 49/440640$. The inequality $3P_x \leq 2D_x$ implies $R_* \leq 1 - 2\delta/3$. Along $\sigma(\ell) = 11/12 - \ell/4$ one obtains exactly

$$E_{\sigma(\ell_0+t)}(h_0 + 3t/2) - E_0 = t(-1/4 + \delta + q + 3R_*/2) \leq 5t/3.$$

For $t = 1/20000$ subtraction yields $49/440640 - 1/12000 = 307/11016000$. This already includes the change of reference exponent $C(\sigma)$; no further deduction of $1/80000$ is appropriate. $\square$

For $1/2 \leq d \leq h$, the slope in (17) is positive, so the endpoint controls the whole interval. The floor, with $\delta = 1/50$, $R = 1$, $q \leq 1/100$, has $E_{\sigma_1}(h) \leq -4327/750000$. For $d_{\min} \leq d \leq 1/2$ the common no-slot bound $R = 76/75 - 2\delta/3$ covers the floor too. Its d slope is $101/150 - \delta/6 > 0$; the geometry variation at fixed d has derivative at most $13/50 + \delta/4 + q \leq 177/200$. Direct substitution gives the uniform bound $E_{\sigma_1}(d) \leq -120907/36000000$. Choose

$$\zeta = m_{\text{ad}}/32 = 307/352512000. \quad (18)$$

It satisfies the supply restriction in Section 5. The slope is at most 2, so extending to $h + \zeta$ costs at most $m_{\text{ad}}/16$ and leaves $307/11750400$. Relative to $C(\beta_*)$ all these exponents decrease further by $\Delta_1 = \beta_* - \sigma_1$. Together with (11) and (12), these estimates cover every nonprincipal physical dyad.

7 Parameter order, height closure, and continuation

Proof of Theorem 1.2. Under the contradiction $\sigma_1 < \beta_* \leq 7/8$, fix $\Delta_1 = \beta_* - \sigma_1 > 0$. All real exponents below are fixed before the target character. First prescribe the low and inverse-normalizer losses so their sum is at most $\Delta_1/2$. Proposition 3.1 then gives, for

$$J_\eta(Z) = I_{\eta, \text{mod}}(Z)/(c_S A_T(Z)), \quad \omega = \Delta_1/2,$$

the bound $|J_\eta(Z)| \ll_\eta Z^{C(\sigma_1) + \omega}$.

For the high side first choose moment losses, strict capacity decrements, the uniform plain mesh and rounding allowances within small fixed portions of m_{ad} . Next choose the fixed even K as in Section 5, then the windows. Only now put $\mu = \sigma_1 \ell / (2K) > 0$. Choose amplitude widths and preliminary prime losses after K , and choose $e > 0$ respecting the detector's bounded presaturation ranges and all local domains. The total central real, dyadic, count and normalizer losses are less than $m_{\text{ad}}/4$. Make the principal losses satisfy

$$(1 + h)e + \varepsilon_{\text{pr}} < m_w/4, \quad e + \varepsilon_{\text{pr}} < m_z/4, \quad e + \varepsilon_{\text{pr}} < \mu/4.$$

Fix (18), then z_∞ in (12); choose the small-row losses below half the margin in (11). The remaining central saving is at least $m_{\text{ad}}/2$. Let

$$m_0 = \min(m_{\text{ad}}, m_w, m_z, \mu, 172249/2000000), \quad m = m_0/4 > 0.$$

All ranges now retain at least $m_0/2$. The earlier mesh was paid by m_{ad} , and is not reselected using the smaller, K -dependent μ . This avoids a circular order of choices.

Choose a target η , fix its final arithmetic data and S , and then fix all internal moment, Sobolev and seminorm orders. The same S is used in every physical sum, identity, calibration, c_S , A_T , H_η and f_η . The preceding estimates give finite exponents A_η, B_η , independent of any later external tail order N , such that

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C(\beta_*) - m} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}. \quad (19)$$

The physical $\text{Im } z$, detector witnesses and prime frequencies use the original fixed matrix's common cumulative $T_1/2$ allocation. They do not get separate newly enlarged buffers.

Let $\varepsilon_{\text{ht}} > 0$ be the minimum of the finitely many specified detector height allowances, and let $A_{\text{ht}, \eta} \leq A_\eta$ dominate their powers. Set

$$\tau_{0, \eta} = \frac{d_{\min} \varepsilon_{\text{ht}}}{20(1 + A_{\text{ht}, \eta})}.$$

For $T_1 = Z^\tau$ with $0 < \tau \leq \min(d_{\min}/100, \tau_{0, \eta})$ and $U \geq Z^{d_{\min}}$, we have eventually $T_1 \leq U^{1/100}$ and $(1 + T_1)^{A_{\text{ht}, \eta}} \leq U^{\varepsilon_{\text{ht}}/10}$. All literal detector height hypotheses thus hold; their crude

$U^{-189/100+o(1)}T_1^2$ error tends to zero. Small and absolute large rows do not need this buffered cutoff. Increasing N in (19) changes only external test seminorms, not internal orders or the real powers.

Now choose $0 < \tau_\eta \leq \min(d_{\min}/100, \tau_{0,\eta}, m/[4(A_\eta+1)])$, then an integer N_η with $B_\eta - N_\eta\tau_\eta < C(\beta_*) - m/2$, and finally the threshold. It follows that

$$|J_\eta - f_\eta| \ll_\eta Z^{C(\beta_*) - \sigma_{\text{hi}}}, \quad \sigma_{\text{hi}} = m/2.$$

Although τ_η, N_η and constants depend on the target, $\omega, \sigma_{\text{hi}}$ do not. Neither J_η nor f_η contains T_1 in its definition. With $\varepsilon_* = \min(\Delta_1/2, \sigma_{\text{hi}}) > 0$, this gives $f_\eta(Z) \ll_\eta Z^{C(\beta_*) - \varepsilon_*}$ at infinity. At zero, move (9) right to any fixed $B > 2$. The Gaussian and the absolutely convergent reciprocal Euler product give $f_\eta(Z) \ll_{\eta,B} Z^{B-11/16}$ for $0 < Z \leq 1$. Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-C(s)} \frac{dZ}{Z}$$

is locally uniformly convergent and holomorphic for $\text{Re } s > \beta_* - \varepsilon_*$. Writing $Z = e^u$, Fourier inversion on $\text{Re } s = 2$ identifies it with $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$. The established integrability justifies the inversion. The identity theorem extends this identity to $\text{Re } s > 1$.

Since $\beta_* - \varepsilon_* > \sigma_1$ and $|H_\eta| \geq 1/2$, $e^{-(s-5/6)^2} F_\eta/H_\eta$ is a holomorphic continuation of $1/L_F^S(s, \eta)$ to $\text{Re } s > \beta_* - \varepsilon_*$. The definition of the supremum supplies some target with a zero whose real part exceeds $\beta_* - \varepsilon_*$; attainment of β_* is unnecessary. Its reciprocal has a pole, a contradiction. The deleted finite Euler factors are nonzero for $\text{Re } s > 0$. Thus $\beta_* \leq \sigma_1$, proving the primitive Hecke assertion and, by restoring finite Euler factors, the imprimitive assertion.

Finally the imported quadratic transfer gives

$$L_F^S(s, \chi \circ N) = L^S(s, \chi) L^S(s, \chi\chi_{-3}).$$

In $0 < \text{Re } s < 1$ both Dirichlet factors are holomorphic, so their zeros cannot be cancelled by poles. Above 1 their Euler products are nonzero. At $1 + it$, $t \neq 0$, the same product excludes zeros. At 1 the only possible pole-zero pairing is between the principal character and χ_{-3} , but $L(1, \chi_{-3}) = \pi/(3\sqrt{3}) > 0$. This proves the stated Dirichlet conclusion, including zeta. $\square$

A Verification and reproducibility

The derivation was developed in repository notes 441–445 [2]. The exact audit `scripts/hybrid_high_continuation_exact_audit.py` uses rational arithmetic for the geometry, Euler margins, capacity supply, all physical row ranges and the continuous completed-square identity. Its output is `output/hybrid-high-continuation-exact-audit.json`. The polynomial identity in Lemma 6.1, rather than a grid sample, proves the continuous central inequality. The audit also checks 16,728 finite frequency-allocation cases. These calculations check the stated algebra and bookkeeping; they do not certify the imported analytic theorems.

The pinned source canonical SHA-256 is

```
42a5ee0febca59fd1def55cfd6c6808c322ef
4237d7726303ea52f711deac6a3 .
```

The research checkpoint for notes 441–445 is `96fcb27a0b8788ad8a346238843906bce0285479`. Independent source-reading reviews and the final paper build manifest are stored in `reviews/2026-10-07/`. The manifest binds the final source, PDF and review hashes.

No external Lean elaboration, kernel build, or Comparator run was performed for this improvement. The source’s generic results and its whole Hecke seven-eighths theorem are explicit imported inputs. Independent verification of that package and a formalization of the present extension remain separate tasks. This paper confines its mathematical assertion to Theorem 1.2.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis*, September 30, 2026 version, source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, [pinned source](#).
- [2] RH–Weil research project, *Variable-slot low bound, shared contours, actual detector capacities, joint errors, and family continuation*, notes 441–445 and their independent reviews, October 7, 2026, checkpoint `96fcb27a0b8788ad8a346238843906bce0285479`.